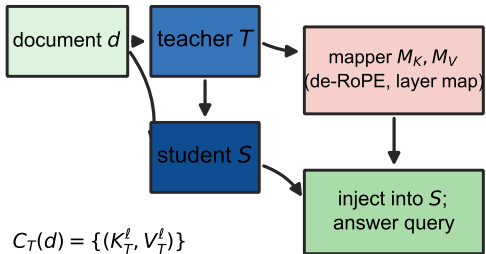


(a) Setting: is the mapped state consumed?



$$C_T(d) = \{(K_T^\ell, V_T^\ell)\}$$

$$C_S(d) = \{(K_S^\ell, V_S^\ell)\}$$

(b) Four-arm injection protocol

